A Finite Mathematical Framework for Exploratory HIV Reservoir-Repair Modeling

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Important limitation. This document reports a synthetic computational concept. It does not establish efficacy or safety, is not a clinical prediction, and provides no treatment, experimental, sequence, delivery, or dosing instructions.

Abstract

We describe an illustrative, bounded model joining sequencing confidence, hypothetical inhibitor-gated editing concepts, conjugate variability, and long-term uncertainty. Feynman-style path weights are used only as an abstract aggregation metaphor; continued fractions impose finite convergence; and elliptic-integral-inspired scaling represents bounded transport heterogeneity. Outputs are normalized indices, not biological repair rates.

Finite Model

For a horizon H and time t, the illustrative index uses a finite exponential trajectory:

$$R(t) = 100\,[1 - \exp(-\alpha t/H)]$$

where $\alpha = A\ q\ E(k)\ C_n$. Here A is a conceptual activity coefficient, q is a data-quality factor, E(k) is a bounded elliptic-scale term, and C_n is a truncated continued fraction. A path-weight summary is finite by construction:

$$Z_N = \Sigma_{j=1..N}\ w_j \exp(-S_j)\ ;\ C_n = a_1 + 1/(a_2 + \ldots + 1/a_n)$$

Statistical Treatment

Illustrative conjugate variability is represented by a beta distribution on [0,1]. The uncertainty band combines scenario dispersion with the square-root horizon factor. This is a visual sensitivity analysis, not a confidence interval from patient data.

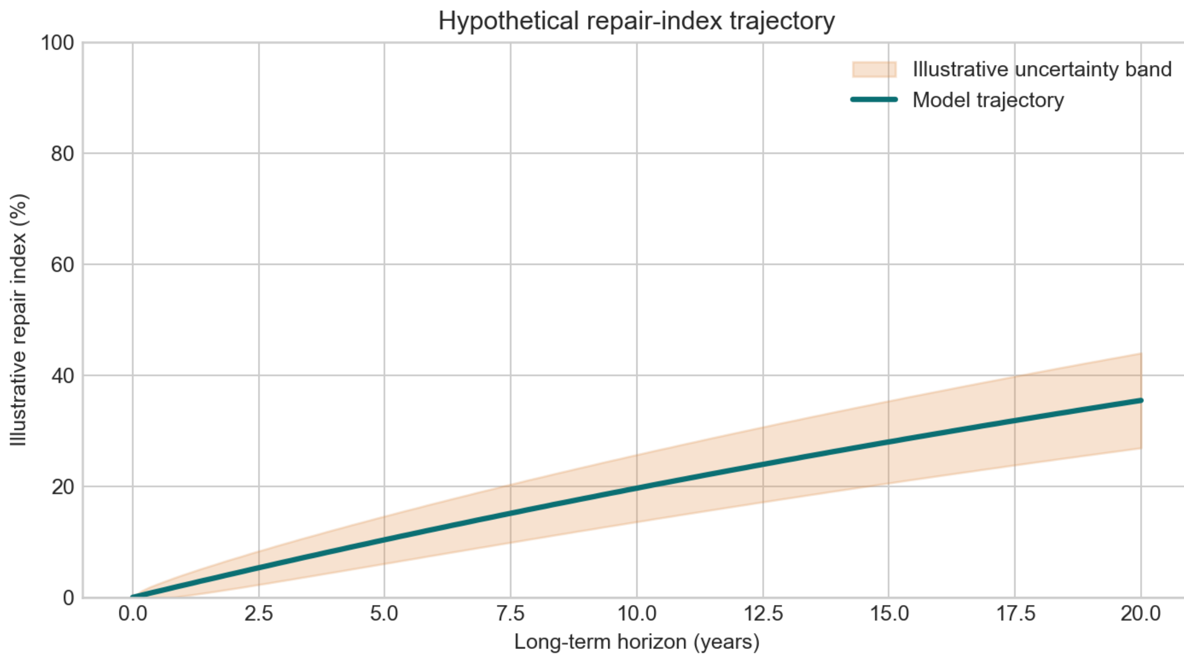

Figure 1 | Illustrative normalized trajectory with synthetic uncertainty envelope.

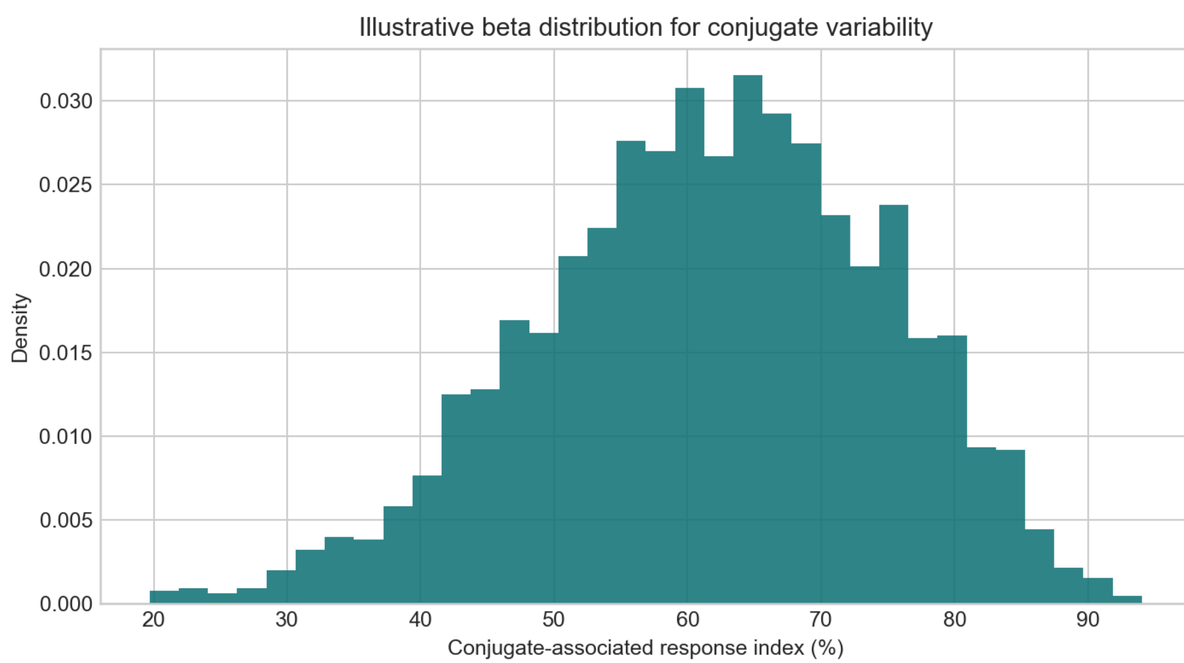

Figure 2 | Synthetic beta-distribution visualization for conjugate-associated variability.

Scenario Characteristics

Component	Representation	Interpretation
Sequencing	q in [0.5, 1.0]	Data quality proxy
Conjugate	Beta-distributed index	Bounded synthetic variability
Horizon	1-50 years	Scenario visualization window
Output	0-100 index	Not a repair percentage or outcome

Discussion

Any potential HIV cure strategy must undergo rigorous preclinical validation, independent replication, clinical trials, regulatory review, and long-term safety monitoring. The model intentionally excludes clinical endpoints and should only be used for mathematical visualization and communication of uncertainty.

References

1. Feynman, R. P. Space-time approach to non-relativistic quantum mechanics. Rev. Mod. Phys. 20, 367-387 (1948).
2. Abramowitz, M. & Stegun, I. Handbook of Mathematical Functions (1972).
3. UNAIDS. Global AIDS Update (accessed 2026).